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binningverdict: a unified decision system for logarithmic versus linear binning of small-angle neutron scattering data

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Abstract

We present **binningverdict**, an open-source Python package for quantitatively selecting logarithmic or linear binning in small-angle neutron scattering (SANS) data reduction. The framework combines correlation-aware variance–bias optimization with Fisher-information analysis in a computationally efficient workflow operating directly on measured (Q, I, σ) profiles. Central to the method is a closed-form reconstruction-error ratio,

$$R_{\text{MSE}} = E_{\log}^* / E_L^*,$$

obtained by extending the Freedman–Diaconis mean-squared-error decomposition to Q -dependent bin widths. The resulting expression depends only on four elementary moments of the data and is entirely model-free. An optional Fisher-information module evaluates finite- N parameter-estimation precision and demonstrates asymptotic scheme-independence between logarithmic and linear discretizations. The package provides convergence diagnostics, batch-processing capability, and millisecond-scale execution using only NumPy. Benchmarks on canonical SANS models, together with applications to measured EQ-SANS datasets, demonstrate robust performance and establish the framework as a practical tool for automated SANS reduction and preprocessing workflows.

PROGRAM SUMMARY

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Program Title: binningverdict

CPC Library link to program files: (to be added by Technical Editor)

Developer's repository link: <https://github.com/cw-do/binningverdict>

Licensing provisions: MIT

Programming language: Python (≥ 3.8)

External routines/libraries: NumPy (≥ 1.20); SciPy (optional, for cylinder form factor in examples)

Nature of problem: Reducing a SANS measurement to an intensity profile $I(Q)$ requires choosing how to discretize the momentum-transfer axis. [Bin width has been treated quantitatively since Mildner and Carpenter \[1\] established the resolution-matched radial bin criterion, and event-mode acquisition now makes the partition freely selectable after the measurement \[2, 3\]. What remains unresolved is the complementary choice of how the bins should be distributed in \$Q\$, which is still made largely by convention.](#) The literature on SAS data information content (Shannon channels [4], Fisher information [5], etc.) characterizes the information ceiling of a measurement but does not recommend a specific binning scheme for a given profile. There is therefore a need for a single, model-free, computationally cheap procedure that takes the measured (Q, I, σ) profile as input and returns a quantitative recommendation.

Solution method: The Freedman–Diaconis variance–bias decomposition is generalized to a Q -dependent bin width $h(Q)$, the resulting reconstruction mean-squared error is evaluated under linear and logarithmic schemes at matched bin count, and the ratio $R_{\text{MSE}} = E_{\text{log}}^*/E_L^*$ is obtained in closed form as $R_{\text{MSE}} = (J_1^\sigma/\bar{\sigma}^2)^{2/3}(J_2/\bar{\beta})^{1/3}$ with four moments computed by trapezoidal quadrature on the supplied input grid. A self-consistency check on a sub-sampled grid flags inputs whose verdict is not yet stable. An optional Fisher branch evaluates a finite- N ratio F_{log}/F_L when a parametric model is supplied. [If the instrumental \$Q\$ -resolution is supplied \(the `dQ` column of a standard reduced file\), a resolution-aware ratio is returned in addition.](#)

Additional comments: The package is single-file, depends only on NumPy, and runs in a few milliseconds on a typical SANS profile. [Both the one-dimensional and the two-dimensional \(annular-average\) Poisson conventions are available, selected by a single keyword.](#) The repository contains worked examples for canonical SANS models, the Fisher branch, and batch analysis of measured profiles, plus a unit test suite.

1. Introduction

The discretization of the momentum-transfer axis is a fundamental step in small-angle neutron scattering (SANS) data reduction because it directly affects reconstruction accuracy, parameter estimation, and downstream analysis workflows [6].

Binning has been studied quantitatively in instrumental analysis for several decades [7, 8]. It is therefore useful to distinguish the part of the problem that is already settled from the part that remains open. Mildner and Carpenter [1] derived the optimization conditions for small-angle scattering resolution and showed that, for azimuthally averaged data, the radial bin width ΔR enters the Q -variance on the same footing as the collimation and wavelength terms; their criterion sets ΔR to a fixed fraction of the detector element size of the optimized instrument and is, in their own words, the most practically useful conclusion of that work. The associated resolution functions were developed further by Pedersen *et al.* [9] and by Barker and Pedersen [10], and the treatment was extended to time-of-flight measurements by Mildner *et al.* [11]. For pulsed-source SANS, Heenan *et al.* [12] gave a resolution expression containing the term $\frac{1}{3}(\Delta R/L_2)^2$ in the squared full width, where ΔR is the width of the annular ring on the detector, alongside the two aperture terms and the wavelength term. In variance this is precisely $h^2/12$ with h the width of the Q bin, so the binning contribution has been carried inside the standard resolution function of the field for four decades. The same work also describes rejecting time channels after measurement to improve resolution at the cost of counting statistics, directly illustrating this trade-off in time-of-flight reduction. Resolution-matched binning has accordingly been routine on both reactor and spallation instruments [13, 12, 14], and facility reduction stacks at ISIS, J-PARC and SNS provide explicit control over the wavelength and radial partitions [15]. This long-standing rule of thumb is therefore well founded. Bins are cut as finely as counting statistics permit, subject to the floor imposed by the instrumental resolution.

That body of work does not, however, address how the bins should be *distributed* along Q once their characteristic width has been fixed. Resolution matching is an instrument-driven and sample-independent criterion; it bounds the bin width from below but says nothing about whether the partition should be uniform in Q or uniform in $\ln Q$. This remaining choice is still made largely by convention and is the focus of the present work. Over the past four decades, several information-theoretic frameworks have charac-

terized the information content of SANS measurements from complementary perspectives. Damaschun *et al.* [16] and Moore [17] introduced the Shannon-channel framework, identifying the number of Shannon channels,

$$N_S = D_{\max}(Q_{\max} - Q_{\min})/\pi,$$

as a sampling-theoretic upper bound on the number of independent intensities recoverable from a scattering profile of finite real-space extent $D_{\max}$, measured over the interval $[Q_{\min}, Q_{\max}]$. Taupin and Luzzati [18] refined the framework for solution scattering. Vestergaard and Hansen [19] introduced the “number of good parameters” N_p as a noise-aware refinement based on Bayesian indirect Fourier transformation, while Rambo and Tainer [20] developed the volume-of-correlation metric V_C . Pedersen *et al.* [21] formulated the problem in terms of Fisher information [5],

$$\mathbb{E}[\mathcal{I}_{ab}(\theta)] \propto \sum_j \frac{\int_{\tilde{Q}_j}^{\tilde{Q}_{j+1}} \partial_a I dQ \int_{\tilde{Q}_j}^{\tilde{Q}_{j+1}} \partial_b I dQ}{\sigma_j^2}, \quad (1)$$

and observed that optimal binning should adapt to regions where the scattering profile varies rapidly. However, they also noted that a practical implementation of Fisher-optimal discretization would be computationally difficult, and the problem was left unresolved. More recent work in the bioSAXS community [22] extended Shannon-based ideas to oversampling-aware fitting strategies, but the specific computational question of logarithmic versus linear sampling of the Q -axis remains unanswered.

A related issue is how much freedom the analyst has in choosing the partition. Recording individual neutron events rather than pre-formed histograms removes the requirement that the partition be fixed at acquisition time, and this capability is not specific to any one facility. Event-mode acquisition has been developed and deployed across the time-of-flight community, and the corresponding reduction methodology, including the statistical treatment of event-based rebinning, has been described in detail by Peterson *et al.* [2, 3] within the Mantid framework [15] that is shared between ISIS and SNS. In practice, this means that the partition can be chosen during reduction on essentially every modern time-of-flight SANS instrument, making an automated selection criterion directly useful.

In recent work [23, 24], we placed bin-width optimization on a quantitative basis by adapting the Freedman–Diaconis (FD) [25] variance–bias

decomposition to SANS data and extending it with Gaussian-process inference to incorporate inter-bin correlations. These studies established scaling relations for optimal bin width under fixed discretization schemes but did not address comparisons between logarithmic and linear sampling. Related work on Bonse–Hart USANS reduction [26] likewise identified the existence of both logarithmic and linear stepping geometries without providing a quantitative selection criterion. The absence of a computationally efficient decision framework motivates the present work.

Here we present `binningverdict`, an open-source Python package that quantitatively determines whether logarithmic, linear, or effectively equivalent binning is preferred for a measured (Q, I, σ) profile. The workflow computes a closed-form reconstruction-error ratio R_{MSE} , together with an optional Fisher-information ratio R_F , and returns a direct binning recommendation. The reconstruction metric extends the FD variance–bias framework to Q -dependent bin widths and reduces the comparison between logarithmic and linear schemes to four elementary moments of the measured profile. The Fisher-information branch further proves asymptotic scheme-independence and characterizes finite- N corrections relevant to practical parameter estimation. The package is implemented entirely in Python using NumPy, supports batch processing and convergence diagnostics, and executes in milliseconds for typical SANS profiles.

The remainder of the paper is organized as follows. Section 2 derives the reconstruction and Fisher-information criteria. Section 3 presents the workflow algorithm and software interface. Section 4 benchmarks the method on representative SANS models and examines the influence of Q -window selection and measurement protocol. Section 5 applies the workflow to measured EQ-SANS datasets. Section 6 relates the present framework to previous information-theoretic approaches, and Section 7 discusses implications for automated SANS reduction workflows. The results show that power-law scattering systematically favors logarithmic binning, whereas non-power-law profiles at wide- Q windows typical of extended-range instruments such as EQ-SANS [27] generally favor linear discretization, with both schemes becoming asymptotically equivalent for parameter estimation at sufficiently large bin counts.

2. Theory

2.1. Reconstruction MSE for Q -dependent bin width

The Freedman–Diaconis decomposition for SANS, derived in Ref. [23] and refined in Ref. [24], expresses the total reconstruction mean-squared error of a binned estimator $I_h(Q)$ as the sum of counting noise and within-bin aliasing distortion. The key observation we exploit here is that this decomposition is local in Q . Applied to a non-uniform partition with local bin width $h(Q)$, the pointwise error density of Refs. [23, 24] generalizes to

$$\varepsilon(Q; h) = \frac{\sigma^2(Q)}{h(Q)} + \frac{h(Q)^2}{12} [I'(Q)]^2 + \mathcal{O}(h^4), \quad (2)$$

which reduces to the FD result of [23] when h is constant and recovers the Poisson limit when $\sigma^2 \propto I$. The total MSE normalized by the window length $L = Q_{\max} - Q_{\min}$ is $E = L^{-1} \int \varepsilon(Q; h) dQ$.

Equation (2) includes only the binning contribution to the Q -variance, not the full instrumental resolution. The term $h^2/12$ is the variance of a uniform distribution of width h and represents the binning contribution to the Q -variance alone. The full instrumental Q -variance of a pinhole small-angle instrument also carries collimation, detector and wavelength terms; in the notation of Mildner and Carpenter [1] the total reads

$$\sigma_Q^2(Q) = \underbrace{B_{\text{geom}} + \frac{1}{12} \left(\frac{\Delta\lambda}{\lambda} \right)^2 Q^2}_{\equiv B(Q)} + \frac{h^2}{12}, \quad (3)$$

where B_{geom} collects the source and sample aperture and detector-element contributions and the second term is the wavelength spread. The equivalent expression for a pulsed source is given by Heenan *et al.* [12], whose annular-ring term reduces to the same $h^2/12$ in variance. For the bin widths used in practice the collimation term is normally the dominant one, so $B(Q)$ cannot be assumed small. On the three measured profiles analysed in Sec. 5 it accounts for between 69 and 79% of the reported ΔQ^2 .

If the reconstruction target is taken to be the true, unsmeared $I(Q)$ rather than the resolution-smeared profile that the instrument actually delivers, then the whole of σ_Q^2 acts on the signal and the error density becomes

$$\varepsilon_{\text{res}}(Q; h) = \frac{\sigma^2(Q)}{h(Q)} + \left[B(Q) + \frac{h(Q)^2}{12} \right] [I'(Q)]^2. \quad (4)$$

Because $B(Q)$ carries no dependence on h , it drops out of the stationarity condition and leaves the optimal bin widths of both schemes unchanged. It contributes instead an additive constant $G \equiv L^{-1} \int B(Q) [I'(Q)]^2 dQ$ that is *identical* in the logarithmic and linear schemes, so that the resolution-aware ratio is

$$R_{\text{MSE}}^{\text{res}} = \frac{E_{\text{log}}^* + G}{E_L^* + G} = \frac{R_{\text{MSE}} + g}{1 + g}, \quad g \equiv G/E_L^* \geq 0. \quad (5)$$

Since $g \geq 0$, adding the resolution term moves the ratio toward unity but cannot move it across unity: $\text{sign}(R_{\text{MSE}}^{\text{res}} - 1) = \text{sign}(R_{\text{MSE}} - 1)$ always. The direction of the verdict is therefore an invariant of the framework, and only its magnitude, and hence the assignment of borderline cases to the tied band, is affected.

A reduced profile does not provide all quantities needed to evaluate g directly. Both G and E_L^* are properties of the underlying profile and of the measurement rather than of the grid the profile happens to be delivered on, so g is independent of that grid. Evaluating g requires the optimal linear-bin width h_L^* , which depends on the underlying noise density of the measurement. The reported uncertainties alone are not sufficient because they are tied to the particular grid on which the data were reduced. Fortunately, g is not needed to choose between logarithmic and linear binning. Equation (5) shows that the resolution term can move the ratio toward unity, but cannot reverse the verdict for any $g \geq 0$.

What can be read directly off a delivered profile is the ratio of the two expected squared variations at that grid,

$$\rho \equiv \frac{\langle B(I')^2 \rangle}{\langle (h_{\text{in}}^2/12) (I')^2 \rangle}, \quad (6)$$

with h_{in} the local spacing of that grid. ρ is dimensionless and invariant under a change of the units of Q , and it answers the practical question of whether the instrument or the binning dominates the definition in Q for the data in hand. Because it carries h_{in}^2 in its denominator, ρ is a property of the grid and not a proxy for g : refining the grid at fixed measurement raises ρ while leaving g untouched. The two are related by $g = (\rho/3)(h_{\text{in}}/h_L^*)^2$, in which the two factors of h_{in} cancel identically, as they must. Appendix A gives the derivation, Sec. 5 reports ρ for the measured profiles, and the package accepts the resolution column of a standard reduced file directly.

2.2. Two canonical schemes and the closed-form ratio

We compare two schemes at fixed total bin count N . The linear scheme uses constant width $h_L = L/N$. The logarithmic scheme uses Q -dependent width $h_{\log}(Q) = Q\Lambda/N$, where $\Lambda = \ln(Q_{\max}/Q_{\min})$.

Each scheme is substituted into the error density Eq. (2). The total MSE in either scheme then takes the familiar variance–bias form $E = \alpha/h + \beta h^2$, with coefficients α and β that depend on the scheme. Minimizing over h inside each scheme gives an optimal error $E^* \propto \alpha^{2/3}\beta^{1/3}$. The ratio of the two optimal errors then has a remarkable property. The window length L , the logarithmic span Λ , and the overall numerical prefactor all cancel out exactly. What remains is a closed-form expression in four moments of the measured profile,

$$R_{\text{MSE}} \equiv \frac{E_{\log}^*}{E_L^*} = \left(\frac{J_1^\sigma}{\overline{\sigma^2}} \right)^{2/3} \left(\frac{J_2}{\bar{\beta}} \right)^{1/3} \quad (7)$$

with the four moments defined as

$$\begin{aligned} \overline{\sigma^2} &\equiv \frac{1}{L} \int \sigma^2 dQ, & J_1^\sigma &\equiv \frac{1}{L} \int \frac{\sigma^2}{Q} dQ, \\ \bar{\beta} &\equiv \frac{1}{L} \int (I')^2 dQ, & J_2 &\equiv \frac{1}{L} \int Q^2 (I')^2 dQ. \end{aligned} \quad (8)$$

A step-by-step derivation is given in Appendix A. $R_{\text{MSE}} < 1$ favors log-binning. $R_{\text{MSE}} > 1$ favors linear binning. The expression is model-free, scale-invariant within a fixed window, and reduces to a window-independent constant for power-law $I(Q) \propto Q^{-\alpha}$ (Sec. 4.1).

When measured uncertainties are supplied, Eq. (7) is used as it stands and no assumption about counting statistics is required. For model-generated profiles a Poisson proxy must be adopted for $\sigma^2(Q)$, and the appropriate choice depends on the detector geometry. For a stepped one-dimensional scan, such as a Bonse–Hart measurement [26], the number of counts accumulated in a bin is proportional to h alone and the proxy is $\sigma^2 \propto I$. For the azimuthal average of a two-dimensional detector the annulus at Q of width h subtends a solid angle proportional to Qh , so that $N_k \propto Q_k h_k I(Q_k)$ and the variance of the intensity estimate scales as $I(Q)/(Qh)$, giving the proxy $\sigma^2 \propto I/Q$. Both conventions are reported throughout Sec. 4; they do not change the sign of $R_{\text{MSE}} - 1$ for any profile examined here, but the two-dimensional weighting amplifies the linear-bin advantage of non-power-law profiles and compresses the log-bin advantage of power-law profiles.

2.3. Fisher information under the two schemes

Pedersen *et al.* [21] derived the per-bin Fisher information contribution Eq. (1) but did not analyze its dependence on the binning scheme. We carry out that analysis now. Under Poisson counting the expected number of counts collected in bin k is set by the solid angle that the bin subtends. For the azimuthal average of a two-dimensional detector the annulus at Q_k of width h_k has area proportional to $Q_k h_k$, so that

$$N_k \propto \Delta\Omega_k I(Q_k) \propto Q_k h_k I(Q_k), \quad (9)$$

and the variance of the corresponding intensity estimate is $\text{Var}(\hat{I}_k) \propto I(Q_k)/(Q_k h_k)$. This weighting is not an idealization. It is what the reduction pipeline of the instrument used here actually applies, each detected neutron being assigned to its Q bin and weighted by the detector solid angle corresponding to that bin [14], and it is the reason the measured uncertainties of Sec. 5 already carry the factor. The Fisher information for any structural parameter θ then becomes a Riemann sum [28]

$$F(\theta) \propto \sum_{k=1}^N \frac{[\partial_\theta I(Q_k)]^2}{I(Q_k)} Q_k h_k \quad (10)$$

for a two-dimensional annular average, and reduces to the same expression without the factor Q_k for a stepped one-dimensional scan. In the continuum limit this converges to $F_\infty = \int w(Q) [\partial_\theta I]^2 / I dQ$ with $w(Q) = Q$ or $w(Q) = 1$ according to the geometry, which depends only on the integrand and not on how Q is discretized. The geometric weight changes the integrand but not the structure of the sum, so the conclusion below is unaffected by it. Therefore

$$\boxed{\lim_{N \rightarrow \infty} \frac{F_{\log}(\theta)}{F_L(\theta)} = 1 \quad \text{for any } \theta.} \quad (11)$$

The structural difference between the MSE and Fisher cases can be made explicit by comparing the per-bin contributions side by side. The reconstruction MSE in bin k centered at Q_k is, from Eq. (2) integrated across the bin,

$$\Delta E_k \simeq \underbrace{\frac{\sigma^2(Q_k)}{h_k}}_{\text{counting}} + \underbrace{\frac{h_k^2}{12} [I'(Q_k)]^2}_{\text{aliasing}} \equiv \frac{\alpha_k}{h_k} + \beta_k h_k^2, \quad (12)$$

which contains the bin width h_k in two non-cancelling roles. The counting term scales as h_k^{-1} , so making bins narrower (more bins total) reduces per-bin variance. The aliasing term scales as h_k^2 , so making bins wider blurs the underlying $I(Q)$. The two roles compete, and the optimum at fixed total bin count depends on the scheme.

By contrast, the per-bin Fisher contribution under Poisson counting is

$$\Delta F_k = \frac{[\partial_\theta I(Q_k)]^2}{\text{Var}(\hat{I}_k)} = \frac{[\partial_\theta I(Q_k)]^2}{I(Q_k)} Q_k h_k. \quad (13)$$

The single equality uses $\text{Var}(\hat{I}_k) \propto I(Q_k)/(Q_k h_k)$ from Eq. (9). A single power of h_k survives at the end, multiplied by the fixed geometric weight Q_k that belongs to the integrand rather than to the partition.

The remaining factor of h_k has a simple interpretation. It does not act as a free knob that the experimenter can tune in either direction. It acts as the local width of the partition that approximates the underlying integral. Recall how a Riemann sum is constructed. To approximate $\int g(Q) dQ$ on a grid $\{Q_k\}$, one writes $\sum_k g(Q_k) \Delta Q_k$, where ΔQ_k is the local spacing of the grid. As the grid is refined, ΔQ_k shrinks at every k and the sum converges to the integral. The limiting integral does not depend on how the grid is refined. A uniform refinement and a non-uniform refinement of the same window give the same answer.

Eq. (13) has exactly the structure of a Riemann sum. The quantity $[\partial_\theta I(Q_k)]^2/I(Q_k)$ plays the role of the integrand, while h_k represents the local interval width ΔQ_k . Summing over k and taking the limit $N \rightarrow \infty$ therefore yields

$$F_\infty = \int \frac{[\partial_\theta I(Q)]^2}{I(Q)} dQ.$$

Whether the discretization is logarithmic or linear, both schemes converge to the same continuum integral. This is the content of Eq. (11).

The MSE in Eq. (12) has no analogous structure. The counting term has h_k in the denominator, the aliasing term has h_k squared in the numerator, and these two scalings cannot be combined into a single Riemann measure. Refining the grid does not push the MSE into a partition-independent limit. Instead, it traces out a variance–bias curve whose minimum location and value depend on how the grid is laid down.

Thus, the two opposing effects of refining the binning—smaller variance per bin against more bins—trade off against each other in the Fisher case

and exactly cancel in the continuum limit. In the MSE case the two effects come from physically distinct sources, namely statistical noise and within-bin signal blur, and their competition does not vanish. Reconstruction sees the scheme. Parameter information does not, asymptotically.

The finite- N residual to Eq. (11) is dictated by the trapezoidal-rule convergence rate of each scheme on the integrand $(\partial_\theta I)^2/I$. This is characterized numerically in Sec. 4.6.

3. Workflow and software

The theory of Sec. 2 is converted into an experimental workflow as Algorithm 1.

Algorithm 1 Log-vs-linear binning recommendation

- 1: **Input:** measured profile (Q, I, σ) ; optional parametric model $I(Q; \theta), \partial_\theta I(Q; \theta)$ at current best θ ; anticipated bin count N .
 - 2: Estimate $I'(Q)$ from (Q, I) by central finite differences.
 - 3: Compute the four moments of Eq. (8) by trapezoidal quadrature.
 - 4: Evaluate R_{MSE} via Eq. (7).
 - 5: Repeat steps 2–4 on a sub-sampled grid (every other point); flag a convergence warning if the two values disagree by more than 10%.
 - 6: **if** parametric model provided **then**
 - 7: Compute $F_L, F_{\log}$ via Eq. (10) at N bins.
 - 8: $R_F \leftarrow F_{\log}/F_L$.
 - 9: **end if**
 - 10: **Verdict:** “log” if $R_{\text{MSE}} < 0.90$; “linear” if $R_{\text{MSE}} > 1.10$; “tied” otherwise.
-

The workflow is implemented in the open-source Python package `binningverdict`, distributed under the MIT license at <https://github.com/cw-do/binningverdict>. The package depends only on NumPy. The user-facing API consists of a single function:

```
from binningverdict import analyze_binning
result = analyze_binning(Q, I, err=sigma)
print(result['verdict'])      # 'linear' / 'log' / 'tied'
print(result['R_MSE'])       # numerical ratio
```

A parametric model can optionally be supplied to also evaluate the Fisher ratio. The result dictionary additionally reports the sub-sampled R_{MSE} used for

Table 1: R_{MSE} for power-law $I(Q) = Q^{-\alpha}$. Values are given for both Poisson conventions: $\sigma^2 \propto I$ for a stepped one-dimensional scan and $\sigma^2 \propto I/Q$ for the azimuthal average of a two-dimensional detector. Both lie below unity at every α and both are window-independent.

α	$R_{\text{MSE}} (\sigma^2 \propto I)$	$R_{\text{MSE}} (\sigma^2 \propto I/Q)$
1.0	0.426	0.909
2.0	0.748	0.905
3.0	0.854	0.923
4.0	0.898	0.937

self-consistency checking, a Boolean **converged** flag, and a human-readable rationale string. The repository contains worked examples for canonical models, the Fisher branch, and batch analysis of measured profiles, plus a unit test suite.

4. Benchmark results

We validate the workflow on representative SANS profiles. Unless noted, the reference window is $[Q_{\min}, Q_{\max}] = [10^{-3}, 0.5] \text{ \AA}^{-1}$, sampled at 10^4 log-spaced points.

4.1. Power-law scattering

For $I(Q) = AQ^{-\alpha}$, the profile is scale-invariant and R_{MSE} is window-independent. Table 1 reports the Poisson-limit values; $R_{\text{MSE}} < 1$ at every α , with $R_{\text{MSE}} \rightarrow 1$ from below as $\alpha \rightarrow \infty$. Power-law scattering is the unique scale-invariant regime in which log-binning is strictly preferred at any window. Adopting the two-dimensional annular weighting $\sigma^2 \propto I/Q$ raises the ratios toward unity but leaves both conclusions intact: every value remains below unity, and the window-independence that follows from scale invariance is exact in both conventions. The practical reading is that log-binning remains the preferred scheme for power-law scattering, but the margin over linear binning is modest once the solid-angle weighting of an area detector is taken into account.

4.2. Non-power-law profiles

Table 2 collects R_{MSE} values for Guinier, sphere, cylinder, Lorentzian peak, and Teubner–Strey profiles at the reference window. All are above

Table 2: R_{MSE} for non-power-law profiles at the reference window, for both Poisson conventions. The Lorentzian peak is parameterized as $I = [(Q - Q^*)^2 + \Gamma^2]^{-1}$ with Q^* and Γ in $\text{\AA}^{-1}$. The Teubner–Strey form is $I = (a + bQ^2 + cQ^4)^{-1}$ with a dimensionless and b, c in $\text{\AA}^2$ and $\text{\AA}^4$ respectively, so that the denominator is dimensionless.

Family	Parameters	$\sigma^2 \propto I$	$\sigma^2 \propto I/Q$
Guinier	$R_g = 30 \text{ \AA}$	2.42	5.49
Guinier	$R_g = 100 \text{ \AA}$	1.93	3.03
Sphere	$R = 30 \text{ \AA}$	2.57	6.24
Sphere	$R = 100 \text{ \AA}$	2.07	3.42
Cylinder	$R=20 \text{ \AA}, L=200 \text{ \AA}$	1.86	3.98
Cylinder	$R=50 \text{ \AA}, L=1000 \text{ \AA}$	1.15	1.81
Lorentzian peak	$Q^*=0.05 \text{ \AA}^{-1}, \Gamma=0.01 \text{ \AA}^{-1}$	1.16	2.48
Teubner–Strey	$a=1, b= -100 \text{ \AA}^2, c=10^5 \text{ \AA}^4$	2.39	5.82

unity, indicating linear-bin advantage. The magnitude is largest for compact objects such as spheres, which concentrate intensity at low Q . The two-dimensional annular weighting roughly doubles every entry, so the linear-bin advantage of non-power-law profiles is strengthened rather than weakened when the solid-angle factor is included.

4.3. Dependence on the measurement window

Figure 1 maps out how R_{MSE} varies with Q_{min} and Q_{max} . Power-law curves are strictly horizontal, confirming scale invariance. All non-power-law profiles show $R_{\text{MSE}} \rightarrow 1$ as the window narrows; the linear-bin advantage is largest at extended- Q operation.

Table 3 summarizes R_{MSE} at three representative low- Q cutoffs.

4.4. Effect of measurement protocol

The generalized form (7) uses measured $\sigma(Q)$ rather than the Poisson proxy $\sigma^2 \propto I$. Table 4 shows how R_{MSE} for a fixed Guinier profile responds to four noise scenarios. The verdict is stable for signal-dominated measurements, weakens with low- Q -weighted counting time, and can flip under fully background-dominated acquisition where σ^2 becomes constant. The last row is the practically important one and is the basis of the trimming guideline given in Sec. 7: a region in which σ has become independent of I carries essentially no structural signal, yet it still enters all four moments of Eq. (8) and can dominate them.

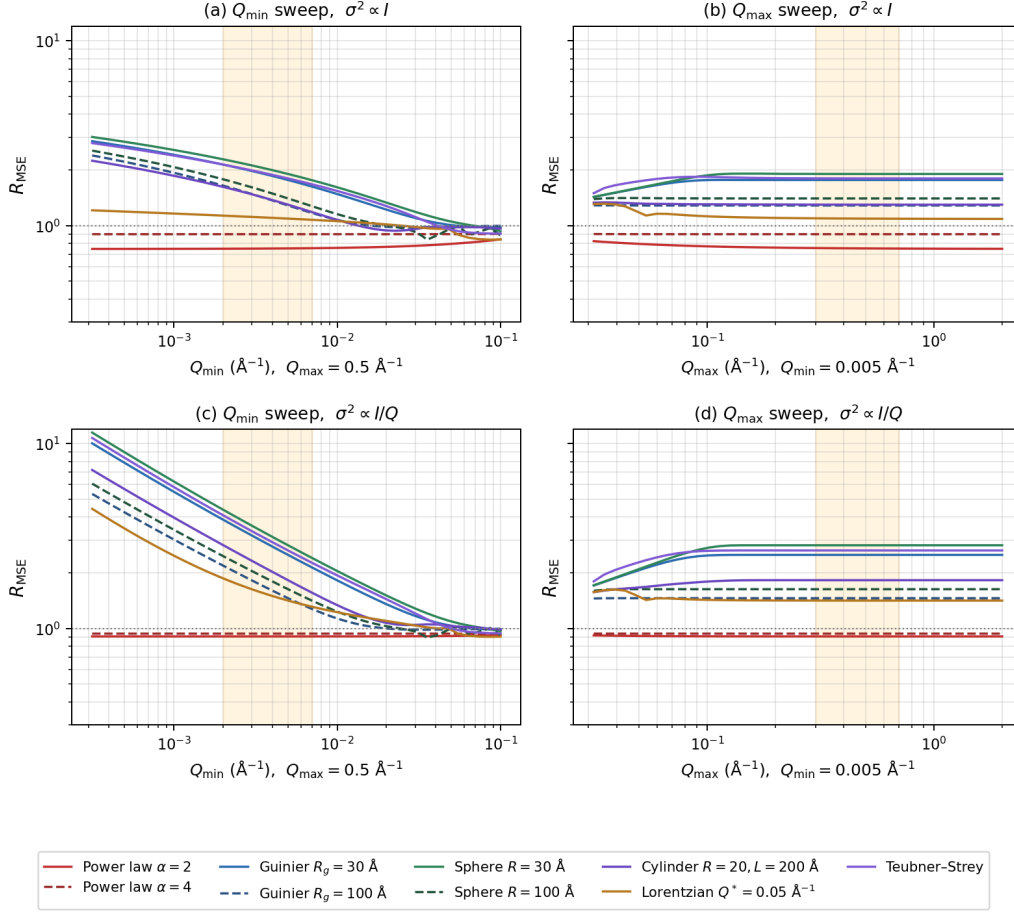


Figure 1: Window dependence of R_{MSE} across nine representative profiles. The top row uses the one-dimensional Poisson convention $\sigma^2 \propto I$ and the bottom row the two-dimensional annular-average convention $\sigma^2 \propto I/Q$. (a), (c) Q_{min} sweep at fixed $Q_{\text{max}} = 0.5 \text{ \AA}^{-1}$, with the shaded band marking the typical EQ-SANS low- Q range. (b), (d) Q_{max} sweep at fixed $Q_{\text{min}} = 0.005 \text{ \AA}^{-1}$. Power-law profiles are scale-invariant in both conventions. All non-power-law profiles approach $R_{\text{MSE}} = 1$ as the window narrows.

4.5. Input quadrature convergence

Figure 2 characterizes how the workflow’s verdict converges with the input bin count N_{in} . For non-power-law profiles (Guinier, sphere, Lorentzian, Teubner–Strey) both schemes agree to within 3% by $N_{\text{in}} = 50$ and the verdict is stable across input schemes. Power-law profiles are the slowest-converging case, and the self-consistency check in Algorithm 1 flags this condition au-

Table 3: R_{MSE} at three representative low- Q cutoffs, $Q_{\text{max}} = 0.5 \text{ \AA}^{-1}$, for both Poisson conventions. The narrowing of the window drives every non-power-law profile toward unity in both conventions, and the power-law entries remain window-independent.

Model	$Q_{\text{min}}=0.002$		$Q_{\text{min}}=0.007$		$Q_{\text{min}}=0.05$	
	I	I/Q	I	I/Q	I	I/Q
Power law $\alpha = 2$	0.75	0.90	0.75	0.90	0.80	0.91
Power law $\alpha = 4$	0.90	0.94	0.90	0.94	0.90	0.94
Guinier $R_g = 30$	2.14	3.88	1.63	2.14	1.00	1.03
Sphere $R = 30$	2.28	4.39	1.76	2.41	1.05	1.09
Cylinder $R=20, L=200$	1.62	2.83	1.18	1.56	0.98	1.03
Lorentzian peak	1.13	1.87	1.07	1.31	0.94	0.98
Teubner–Strey	2.14	4.10	1.67	2.27	0.95	1.00

Table 4: Sensitivity of R_{MSE} to the measurement noise model. Guinier profile, $R_g = 30 \text{ \AA}$, reference window.

Noise model $\sigma^2(Q)$	R_{MSE}
Poisson, $\sigma^2 = I(Q)$ (baseline)	2.42
Poisson, $\sigma^2 = I(Q)/Q$ (2D annular average)	5.49
Poisson + high- Q background floor	2.41
Q-dependent counting time, low- Q weighted $\times 10$	1.45
Constant σ (background-dominated)	0.73

tomatically. Table 5 quantifies the residual disagreement at $N_{\text{in}} = 100$. The power-law rows show the largest gap between the two input schemes in both Poisson conventions, of order ten to fifteen percent, against one to three percent for the remaining profiles. Whether that gap changes the reported verdict depends on where the values sit relative to the tolerance band rather than on the size of the gap alone, which is why the convergence diagnostic rather than the verdict label is the quantity to monitor.

Two practical recommendations follow. First, supply log-spaced input where possible, since convergence is faster. Second, when the convergence diagnostic flags a power-law-like profile, recompute on a denser input grid by supplying a less aggressively rebinned profile, and continue until the verdict stabilizes.

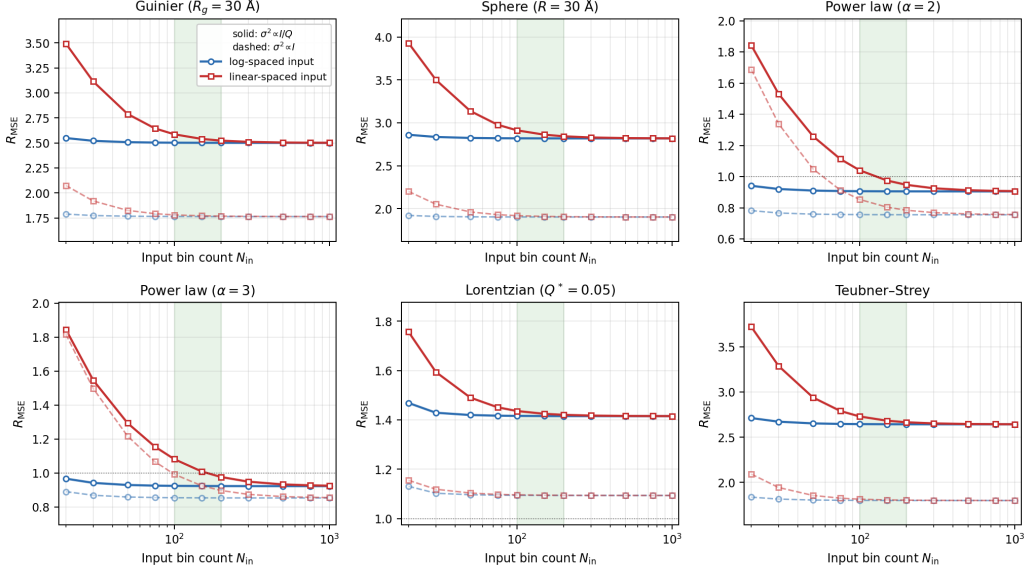


Figure 2: Convergence of the R_{MSE} estimator with input bin count N_{in} for log-spaced (blue circles) and linear-spaced (red squares) input. Solid lines use the two-dimensional convention $\sigma^2 \propto I/Q$ and faint dashed lines the one-dimensional convention $\sigma^2 \propto I$. Both input schemes converge to the same limit in either convention, and for power-law profiles the convergence is slower in the linear-input branch. The shaded band marks $N_{\text{in}} \in [100, 200]$, the typical regime of standard SANS reduction.

4.6. Fisher information: asymptotic equivalence

Figure 3 shows the finite- N Fisher analysis. Panel (a) plots F_{log}/F_L versus N for four representative models. All curves approach unity as N grows, confirming Eq. (11). Panel (b) shows the absolute deviation from the continuum limit. For Guinier, sphere, and Lorentzian profiles, both schemes reach the continuum limit to within 0.1% by $N \approx 100$; parameter precision is scheme-independent at any realistic SANS bin count. Power-law scattering is the exception, with $F_{\text{log}}/F_L \approx 3$ at $N = 20$ in the unweighted case and convergence to within 1% requiring $N \gtrsim 500$. Including the solid-angle weight $w(Q) = Q$ improves this appreciably. The power-law ratio falls from 2.83 to 1.69 at $N = 20$ and from 1.11 to 1.05 at $N = 100$, because the weight suppresses the low- Q end of the integrand, which is where the two partitions differ most. The geometric factor therefore accelerates the approach to Eq. (11) rather than impeding it.

Table 5: Verdict consistency between log- and linear-spaced input schemes at $N_{\text{in}} = 100$ over $[0.005, 0.3] \text{ \AA}^{-1}$, for both Poisson conventions. Verdicts: “L”=linear, “log”=logarithmic, “~”=tied. R_{MSE} values shown in parentheses. The largest numerical disagreement between the two input schemes occurs for the power-law profiles in both conventions. Whether it crosses a verdict boundary depends on where the values happen to sit relative to the tolerance band.

Model	$\sigma^2 \propto I$			$\sigma^2 \propto I/Q$		
	log in	linear in	same?	log in	linear in	same?
Guinier $R_g=30$	L (1.76)	L (1.78)	yes	L (2.50)	L (2.59)	yes
Sphere $R=30$	L (1.90)	L (1.92)	yes	L (2.82)	L (2.91)	yes
Lorentzian peak	~(1.09)	~(1.10)	yes	L (1.42)	L (1.44)	yes
Teubner–Strey	L (1.80)	L (1.81)	yes	L (2.65)	L (2.73)	yes
Power law $\alpha=2$	log (0.76)	log (0.85)	yes	~(0.91)	~(1.04)	yes
Power law $\alpha=3$	log (0.86)	~(0.99)	no	~(0.93)	~(1.08)	yes

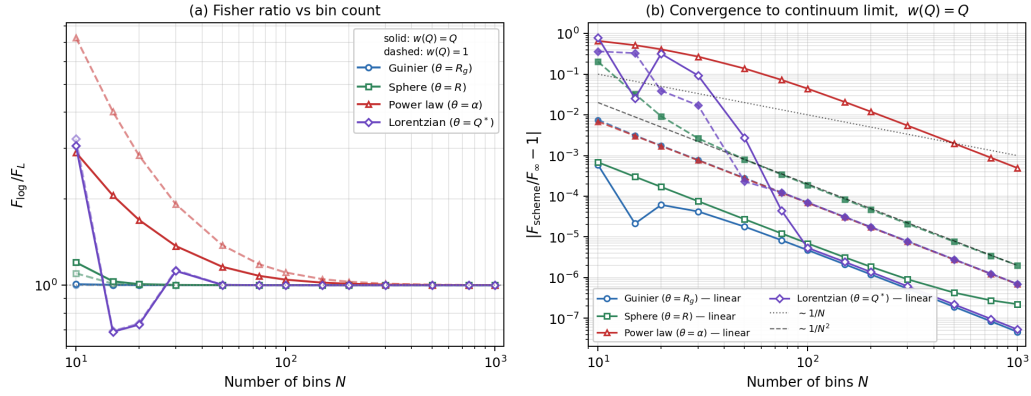


Figure 3: Finite- N Fisher information under linear and log binning. (a) Ratio F_{log}/F_L versus N for four representative models. Solid lines carry the solid-angle weight $w(Q) = Q$ of a two-dimensional annular average and faint dashed lines the unweighted case $w(Q) = 1$. All curves approach unity in the large- N limit, confirming Eq. (11), and the geometric weight accelerates rather than impedes the approach. (b) Convergence of each scheme to the continuum Fisher integral for $w(Q) = Q$. Solid lines mark the linear scheme, dashed lines mark the log scheme. Reference slopes $1/N$ and $1/N^2$ indicate expected Riemann-sum convergence.

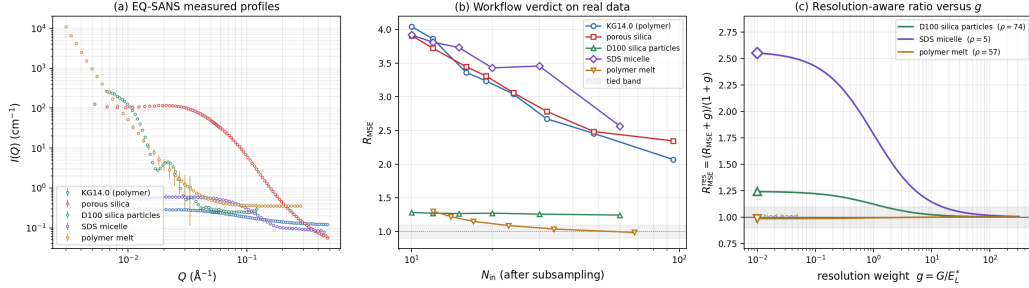


Figure 4: Workflow applied to five measured EQ-SANS profiles. (a) Measured intensity profiles. (b) R_{MSE} as a function of input bin count N_{in} , obtained by subsampling each profile by progressively larger strides. Shaded band marks the tied verdict region. (c) Resolution-aware ratio $R^{\text{res}} = (R_{\text{MSE}} + g)/(1 + g)$ of Eq. (5) as a function of the weight g , for the three profiles that carry a ΔQ column. Each curve starts at the plain ratio at $g \rightarrow 0$, shown by the open symbol, and approaches unity monotonically without ever crossing it, reaching unity only as $g \rightarrow \infty$. The shaded band is the tied region. A profile whose plain ratio is already near the band is carried inside it at modest g , whereas one further away retains its preference over a wide range.

5. Application to measured EQ-SANS data

We close the validation by applying the workflow to five measured EQ-SANS profiles spanning distinct soft-matter and porous-material classes (Fig. 4, Table 6). The first four samples—representing distinct soft-matter classes—all return a linear verdict at the wide- Q EQ-SANS window in which they were measured. R_{MSE} values cover a substantial range (1.24–2.57), with the smallest advantage for D100, whose narrower window $[0.007, 0.12] \text{\AA}^{-1}$ moves the verdict toward the tied regime in line with the trend of Fig. 1. SDS micelles, where peak-like correlation features dominate, give the largest linear-bin advantage. These outcomes confirm the central prediction: at extended- Q operation typical of EQ-SANS, non-power-law profiles benefit measurably from linear binning.

The polymer melt behaves differently because it combines two distinct regimes. At low Q the profile follows a near-Porod $I \propto Q^{-4}$ decay over almost two decades, while at $Q \gtrsim 0.05 \text{\AA}^{-1}$ it transitions to an essentially flat plateau. Restricted analyses on the two regions separately give $R_{\text{MSE}} \approx 0.99$ on the Porod region (consistent with the $\alpha \rightarrow 4$ limit of Table 1) and $R_{\text{MSE}} \approx 0.71$ on the plateau region (where flat $I(Q)$ favors log-binning). Combined over the full window, the two contributions balance to a tied verdict, $R_{\text{MSE}} = 0.99$. The subsampling curve in Fig. 4(b) further shows

Table 6: Workflow verdict on five measured EQ-SANS profiles at full input resolution.

Sample	N_{in}	Q range ($\text{\AA}^{-1}$)	R_{MSE}	Verdict
KG14.0 (polymer)	95	[0.005, 0.48]	2.07	linear
porous silica	95	[0.005, 0.48]	2.34	linear
D100 silica particles	60	[0.007, 0.12]	1.24	linear
SDS micelle	60	[0.012, 0.45]	2.57	linear
polymer melt	68	[0.003, 0.28]	0.99	tied

Table 7: Effect of the instrumental resolution of the three measured profiles that carry a ΔQ column. $B/\Delta Q^2$ is the share of the reported resolution that is collimation and wavelength rather than binning. ρ , Eq. (6), compares the resolution and binning contributions to the expected squared variation at the grid the profile was delivered on. The verdict column is that of the plain ratio; by Eq. (5) the resolution-aware ratio has the same sign of $R - 1$ for any $g \geq 0$, so the direction is unchanged whatever the value of g .

Sample	R_{MSE}	Verdict	$B/\Delta Q^2$	ρ
D100 silica particles	1.24	linear	69%	74.1
SDS micelle	2.57	linear	79%	4.7
polymer melt	0.99	tied	77%	56.5

that this verdict is itself N_{in} -sensitive: at $N_{\text{in}} \lesssim 20$ the verdict drifts into the linear regime, and the convergence diagnostic built into the workflow flags this case automatically.

5.1. Effect of the instrumental resolution term

Three of the five profiles carry the instrumental resolution as a fourth column, which allows Eqs. (5) and (6) to be evaluated directly. The reported ΔQ is the total Q -resolution and therefore already contains the binning contribution of the delivered grid, so the scheme-independent part is recovered as $B(Q) = \Delta Q^2 - h^2/12$ with h the local spacing of that grid. Across the three samples this leaves 69–79% of ΔQ^2 in B , confirming the expectation that the collimation and wavelength terms dominate over the binning term at the grids actually delivered. Table 7 gives the resulting quantities.

The values of ρ confirm the expectation directly. For D100 and the polymer melt the resolution contribution exceeds the binning contribution by factors of 74 and 57, so at the grids these profiles were delivered on it is the

instrument, and not the choice of partition, that sets the definition in Q . The SDS micelle is delivered on a finer grid relative to its resolution and has $\rho = 4.7$, but even there the instrumental term is the larger of the two. In no case is the binning the dominant contribution.

Equation (5) fixes the verdict without requiring a numerical value of g . The resolution-aware ratio approaches unity from whichever side the plain ratio sits on, monotonically in g , and it reaches unity only in the limit $g \rightarrow \infty$, which is the statement that a measurement whose Q -variance is entirely instrumental is indifferent to the binning scheme. Figure 4(c) displays this for the three profiles. A profile whose plain ratio already lies close to the tolerance band, as D100 does at 1.24, will be carried inside it once the resolution term is included, while one further from the boundary, such as the SDS micelle at 2.57, retains its preference over a wide range of g .

For the same reason, the plain ratio is usually the more useful decision statistic in practice. The resolution-aware ratio tends to unity for every sample as g grows, so it progressively stops discriminating, and it does so because it is then reporting a property of the instrument rather than of the one choice under the analyst’s control. We therefore recommend the plain ratio when the reduced profile is the deliverable, which is the usual situation in SANS reduction, and the resolution-aware form when the downstream target is an estimate of the unsmeared $I(Q)$.

6. Relation to prior information-theoretic frameworks

The present work builds on earlier analyses of SAS information content rather than replacing them. Table 8 summarizes.

The Shannon-channel framework [4, 16, 17, 18] sets a sampling-theoretic upper bound on the number of independent intensities recoverable from a profile of finite real-space extent, but is silent on how those intensities should be distributed in Q . N_p [19] and V_C [20] refine the information bound by accounting for noise and correlation, but again make no recommendation about the distribution of bins. Pedersen *et al.* [21] did pose the binning question through the Fisher matrix Eq. (1) but explicitly noted that translating it into a practical algorithm would be “somewhat complicated” and left it as an open direction. The present work follows that direction by restricting attention to the two practically relevant canonical schemes (log and linear) rather than the full optimization, which permits the closed form Eq. (7) and removes the model dependence in the MSE branch. The Fisher branch sharpens Pedersen’s observation by showing asymptotic scheme-independence, a

Table 8: Relation of the present work to prior frameworks.

Reference	Quantity	Question addressed
Damaschun [16], Moore [17]	Shannon channels N_S	Maximum independent intensities for given $D_{\max}$
Taupin & Luzzati [18]	Refined N_S	Solution-scattering specifics
Vestergaard & Hansen [19]	Number of good params N_p	Noise-aware effective dimensionality
Rambo & Tainer [20]	Volume of correlation V_C	Mass and resolution from oversampled data
Pedersen [21]	Fisher matrix Eq. (1)	Optimal binning in principle (not pursued)
Tung <i>et al.</i> [23, 24]	FD/GP optimal h at fixed scheme	Bin- <i>width</i> selection within a scheme
Present work	R_{MSE} , asymptotic $F_{\log}/F_L$	Cross- <i>scheme</i> comparison: log vs. linear

conclusion that was not articulated in [21].

Our recent FD and GP-based analyses [23, 24] provide the variance–bias decomposition of which Eq. (2) is the Q -dependent extension. Those works answered “how wide should the bin be?” The present work answers “how should the bins be distributed?” The two layers compose. The user picks the scheme by R_{MSE} and the within-scheme bin count by FD/GP.

7. Discussion

The unified verdict is window-dependent and operationally simple. Power-law profiles favor log-binning at every Q window. Non-power-law profiles favor linear at extended- Q windows characteristic of EQ-SANS-class instruments, with ratios of 1.5–2.3 in the Poisson limit. At narrow windows the schemes converge to statistical equivalence.

The practical implication splits along the use case. When the reduced profile $I(Q)$ feeds further reconstruction-stage processing such as multiple-scattering correction, desmearing of USANS profiles [26], indirect Fourier transform, Kratky displays, or model-free data releases, the binning choice has direct consequences. The effect is largest at the wide Q -windows where

modern instruments operate. When the only target is parameter estimation through structural model fitting, asymptotic scheme-independence makes the choice immaterial at typical bin counts.

7.1. Relation to resolution-matched binning

Conventional practice fixes the bin width by matching it to the instrumental resolution, and it is worth stating precisely how that criterion and the present one relate, because they answer different questions and do not compete. Equation (3) makes the comparison concrete. Resolution matching is the statement that there is no point in making h so small that $h^2/12 \ll B(Q)$, because the Q -variance is then already dominated by collimation and wavelength spread and further subdivision buys no additional definition in Q . It is an instrument-driven criterion, it depends on aperture sizes, flight paths and $\Delta\lambda/\lambda$ rather than on the sample, and it acts as a lower bound on the useful bin width.

The variance–bias criterion answers the complementary question of how the counting statistics of a particular sample should be traded against the loss of definition within a bin. It is sample-driven, it depends on $I(Q)$ and $\sigma(Q)$ rather than on the instrument, and it selects a width, and now a distribution, within the range that resolution matching leaves open. The two criteria therefore address opposite limits of the problem. Resolution matching says how fine is too fine; the variance–bias analysis says how coarse is too coarse and, through R_{MSE} , where the bins should be placed. Their independence follows directly from the result in Sec. 2: the resolution term enters the error density without any dependence on h , so it leaves both the optimal widths and the direction of the log-versus-linear verdict untouched, and modifies only the magnitude of the advantage. A user who has already applied resolution matching to fix the bin count can apply the present criterion on top of it without revisiting that choice.

7.2. Practical guidelines

The sensitivity study in Sec. 4.4 suggests a few practical rules. The one case in which the verdict is genuinely fragile is a profile whose high- Q end has become dominated by incoherent or instrumental background. There the intensity is flat, the uncertainty stops tracking the intensity, and σ^2 tends to a constant; the last row of Table 4 shows that this alone can move a Guinier profile from a clear linear verdict to a log verdict. Such a region carries no

structural information but still contributes to all four moments of Eq. (8), and because those moments are Q -weighted it contributes disproportionately.

We therefore recommend the following. First, trim the profile to the range over which the sample signal exceeds the background before submitting it to the workflow; the range used for model fitting is normally the right choice. Second, use the local logarithmic slope $d \ln \sigma^2 / d \ln I$ as a diagnostic, since a value near unity indicates a signal-dominated region and a value near zero indicates that σ has decoupled from I and the region should be trimmed. Third, when trimming is ambiguous, run the workflow on both the trimmed and untrimmed profiles; agreement indicates that the verdict is not driven by the background region, and disagreement should be resolved in favour of the trimmed result. Fourth, retain the convergence diagnostic of Algorithm 1 in all cases, since a profile that is too coarsely sampled produces an unstable verdict independently of any background issue. These checks are inexpensive and are reported automatically by the package.

7.3. Applicability beyond SANS

The derivation is not specific to neutron scattering. The workflow consumes a radially averaged (Q, I, σ) profile and, optionally, a resolution column, and returns a verdict; the radiation used to obtain that profile does not enter. The framework therefore applies unchanged to small-angle X-ray scattering, where the fine pixellation of modern area detectors makes the partition at least as free a choice as it is in event-mode SANS. Two differences are worth anticipating. The solid-angle weighting of Eq. (9) applies in the same form to an azimuthally averaged two-dimensional SAXS detector, so the $\sigma^2 \propto I/Q$ convention is the appropriate Poisson proxy there as well. The instrumental term $B(Q)$, on the other hand, is generally smaller for a well-collimated synchrotron beam with a narrow energy bandpass than for a SANS instrument, where the aperture terms are large and, on reactor instruments using a velocity selector, the wavelength spread reaches five to thirty percent. The contrast is less pronounced for time-of-flight SANS, whose wavelength resolution is an order of magnitude better than that of a velocity selector [12, 14], so that $B(Q)$ there is dominated by the geometric terms. In either case a smaller B means a smaller g in Eq. (5) and a correspondingly sharper verdict.

A practical complement to fixed-scheme binning is fully adaptive $h(Q)$,

obtained by pointwise minimization of Eq. (2). This yields

$$h^*(Q) \propto \left[\frac{\sigma^2(Q)}{(I'(Q))^2} \right]^{1/3}.$$

Time-of-flight SANS instruments that store individual neutron events permit such post-hoc adaptive rebinning. Event-mode acquisition and the associated reduction methodology have been developed across the time-of-flight community rather than at any single facility [2, 3, 15], and the EQ-SANS instrument used here [27, 14] is one implementation among several. The reduction-stack overhead, however, makes the present fixed-scheme verdict the most immediately actionable result. Inter-bin correlation corrections introduced in Ref. [24] modify the coefficients in Eq. (7) and will be treated in future work.

8. Conclusion

We have presented `binningverdict`, an open-source Python package for quantitative selection between logarithmic and linear binning in SANS data reduction workflows. The framework combines a closed-form reconstruction-error estimator, $R_{\text{MSE}} = E_{\log}^*/E_L^*$, with an optional Fisher-information analysis to provide an automated and computationally inexpensive binning recommendation directly from measured (Q, I, σ) profiles. The reconstruction criterion is obtained by extending the Freedman–Diaconis variance–bias decomposition to Q -dependent bin widths, while the Fisher branch establishes asymptotic scheme-independence for parameter estimation.

Benchmark calculations on canonical scattering models and applications to measured EQ-SANS datasets demonstrate that power-law profiles systematically favor logarithmic binning in a scale-invariant manner, whereas non-power-law profiles at the wide Q -windows typical of extended-range instruments generally favor linear discretization. The workflow further includes a self-consistency diagnostic that automatically identifies input grids for which the verdict has not yet converged.

Because the method requires only elementary quadrature and finite-difference operations, it executes in milliseconds using only NumPy and is suitable for integration into automated SANS and USANS reduction pipelines where reproducible discretization criteria are important for downstream analysis and reconstruction accuracy. The package is distributed under the MIT license and provides batch-processing utilities, worked examples, and an optional

Fisher-analysis interface through the single function call `analyze_binning(Q, I, sigma)`.

Appendix A. Step-by-step derivation of Eq. (7)

This appendix derives the closed-form ratio $R_{\text{MSE}} = E_{\log}^*/E_L^*$ in detail. We start from the Q -dependent error density

$$\varepsilon(Q; h) = \frac{\sigma^2(Q)}{h(Q)} + \frac{h(Q)^2}{12} [I'(Q)]^2, \quad (\text{A.1})$$

identical to Eq. (2) of the main text. The total MSE normalized by the window length $L = Q_{\max} - Q_{\min}$ is

$$E[h] = \frac{1}{L} \int_{Q_{\min}}^{Q_{\max}} \varepsilon(Q; h) dQ. \quad (\text{A.2})$$

Linear scheme

The linear scheme uses constant bin width $h_L = L/N$. Substituting $h(Q) = h_L$ into Eq. (A.1) and integrating gives

$$E_L(h_L) = \frac{1}{h_L} \overline{\sigma^2} + \frac{h_L^2}{12} \bar{\beta}, \quad (\text{A.3})$$

where the moments

$$\overline{\sigma^2} = \frac{1}{L} \int \sigma^2 dQ, \quad \bar{\beta} = \frac{1}{L} \int (I')^2 dQ \quad (\text{A.4})$$

are simple Q -averages of σ^2 and $(I')^2$.

Logarithmic scheme

The logarithmic scheme uses the Q -dependent width $h_{\log}(Q) = Q \delta$, with $\delta = \Lambda/N$ and $\Lambda = \ln(Q_{\max}/Q_{\min})$. Substituting into Eq. (A.1) gives

$$\varepsilon_{\log}(Q; \delta) = \frac{\sigma^2(Q)}{Q \delta} + \frac{Q^2 \delta^2}{12} [I'(Q)]^2. \quad (\text{A.5})$$

Integrating over Q and dividing by L yields

$$E_{\log}(\delta) = \frac{1}{\delta} J_1^\sigma + \frac{\delta^2}{12} J_2, \quad (\text{A.6})$$

where the new moments

$$J_1^\sigma = \frac{1}{L} \int \frac{\sigma^2}{Q} dQ, \quad J_2 = \frac{1}{L} \int Q^2 (I')^2 dQ \quad (\text{A.7})$$

arise from the explicit factors of Q^{-1} and Q^2 produced by substituting $h_{\log}(Q) = Q\delta$.

Optimal errors in each scheme

Both Eq. (A.3) and Eq. (A.6) have the same generic form

$$E(x) = \frac{a}{x} + b x^2, \quad (\text{A.8})$$

for a width parameter x and positive coefficients a, b . The minimum is found by setting $dE/dx = -a/x^2 + 2bx = 0$, giving

$$x^* = \left(\frac{a}{2b}\right)^{1/3}, \quad E^* = \frac{3}{2^{2/3}} a^{2/3} b^{1/3}. \quad (\text{A.9})$$

Applied to the linear scheme with $a = \overline{\sigma^2}$ and $b = \bar{\beta}/12$ yields

$$h_L^* = \left(\frac{6\overline{\sigma^2}}{\bar{\beta}}\right)^{1/3}, \quad E_L^* = K \overline{\sigma^2}^{2/3} \bar{\beta}^{1/3}, \quad (\text{A.10})$$

with $K = (3/2^{2/3}) 12^{-1/3}$. Applied to the logarithmic scheme with $a = J_1^\sigma$ and $b = J_2/12$ yields

$$\delta^* = \left(\frac{6J_1^\sigma}{J_2}\right)^{1/3}, \quad E_{\log}^* = K (J_1^\sigma)^{2/3} J_2^{1/3}. \quad (\text{A.11})$$

Cancellation in the ratio

Forming the ratio $E_{\log}^*/E_L^*$, the prefactor K cancels exactly. The window length L that appears implicitly through the moment definitions also cancels, since both numerator and denominator moments are normalized by L . The logarithmic span Λ does not appear at all in the optimal errors, only in the relation between δ^* and the actual bin count $N = \Lambda/\delta^* = N(J_1^\sigma, J_2, \Lambda)$, and this Λ does not propagate into $E_{\log}^*$ itself. The remaining expression is

$$R_{\text{MSE}} = \frac{E_{\log}^*}{E_L^*} = \left(\frac{J_1^\sigma}{\overline{\sigma^2}}\right)^{2/3} \left(\frac{J_2}{\bar{\beta}}\right)^{1/3}, \quad (\text{A.12})$$

which is Eq. (7) of the main text.

Poisson special case

When $\sigma^2(Q) = cI(Q)$ for a positive constant c (the Poisson limit), the constant c appears in the same way in $\overline{\sigma^2}$ and in J_1^σ , and it cancels in their ratio. The expression then reduces to

$$R_{\text{MSE}}^{\text{Poisson}} = \left(\frac{J_1}{\bar{I}} \right)^{2/3} \left(\frac{J_2}{\bar{\beta}} \right)^{1/3}, \quad (\text{A.13})$$

with $\bar{I} = L^{-1} \int I dQ$ and $J_1 = L^{-1} \int I/Q dQ$. The Poisson-limit form is what we use for all model-generated benchmarks in Sec. 4.

Appendix B. Resolution-aware form of the ratio

This appendix derives Eq. (5) and establishes the invariance of the verdict under the instrumental resolution term.

The measured intensity is a convolution of the true profile with the instrumental resolution function. A random displacement of the nominal Q by an amount of variance σ_Q^2 produces, to leading order in the displacement, an *expected squared variation*

$$\langle [I(Q + \delta Q) - I(Q)]^2 \rangle \simeq \sigma_Q^2 [I'(Q)]^2. \quad (\text{B.1})$$

This quantity should be distinguished from the convolution bias. The bias is the difference between the true and the smeared intensity, $\langle I(Q + \delta Q) \rangle - I(Q) \simeq \frac{1}{2} \sigma_Q^2 I''(Q)$, whose square is of order $\sigma_Q^4 [I'']^2$ and whose leading contribution comes from the curvature rather than the slope. The quantity that belongs alongside the $h^2/12$ term of Eq. (2) is Eq. (B.1), because that term is itself the expected squared variation of I across a bin, namely $\langle [I(Q) - \bar{I}_{\text{bin}}]^2 \rangle \simeq (h^2/12) [I'(Q)]^2$. The two displacement sources, instrumental smearing and within-bin averaging, are independent, so their variances add. That is exactly why σ_Q^2 is written as a sum in Eq. (3) and why the two terms appear inside a single bracket below. With $\sigma_Q^2 = B(Q) + h^2/12$ the error density of Eq. (2) becomes Eq. (4),

$$\varepsilon_{\text{res}}(Q; h) = \frac{\sigma^2(Q)}{h(Q)} + \left[B(Q) + \frac{h(Q)^2}{12} \right] [I'(Q)]^2. \quad (\text{B.2})$$

Substituting the linear scheme $h = h_L$ and integrating gives

$$E_L^{\text{res}}(h_L) = \frac{\overline{\sigma^2}}{h_L} + \frac{h_L^2}{12} \bar{\beta} + G, \quad G \equiv \frac{1}{L} \int B(Q) [I'(Q)]^2 dQ. \quad (\text{B.3})$$

Substituting the logarithmic scheme $h = Q\delta$ gives

$$E_{\log}^{\text{res}}(\delta) = \frac{J_1^\sigma}{\delta} + \frac{\delta^2}{12} J_2 + G. \quad (\text{B.4})$$

The same constant G appears in both, because $B(Q)$ multiplies $[I'(Q)]^2$ without reference to the partition and is therefore integrated identically whichever scheme is used.

Since G carries no dependence on h_L or on δ , the stationarity conditions $dE/dh = 0$ are unchanged, and the optimal widths remain those of Eqs. (A.10) and (A.11). Evaluating at the optima,

$$R_{\text{MSE}}^{\text{res}} = \frac{E_{\log}^* + G}{E_L^* + G} = \frac{R_{\text{MSE}} + g}{1 + g}, \quad g \equiv \frac{G}{E_L^*} \geq 0. \quad (\text{B.5})$$

Two consequences follow. First, differentiating with respect to g gives

$$\frac{\partial R_{\text{MSE}}^{\text{res}}}{\partial g} = \frac{1 - R_{\text{MSE}}}{(1 + g)^2}, \quad (\text{B.6})$$

so the ratio moves monotonically toward unity as the resolution term grows in relative importance, from above if $R_{\text{MSE}} > 1$ and from below if $R_{\text{MSE}} < 1$. Second, and more importantly,

$$R_{\text{MSE}}^{\text{res}} - 1 = \frac{R_{\text{MSE}} - 1}{1 + g}, \quad (\text{B.7})$$

so that $R_{\text{MSE}}^{\text{res}} - 1$ and $R_{\text{MSE}} - 1$ have the same sign for every finite $g \geq 0$, whatever the value of g and therefore without any need to determine it. The resolution term can therefore never reverse the log-versus-linear verdict. It can only reduce the margin, which in the limit $g \rightarrow \infty$ drives the ratio to exactly unity, the statement that a measurement whose Q -variance is entirely instrumental is indifferent to the binning scheme. This is the expected behaviour and is the reason the plain ratio R_{MSE} remains the appropriate default when the deliverable of the reduction is the resolution-smeared profile together with its ΔQ table, as is standard practice in SANS.

What g depends on, and what ρ measures

At the optimum the two terms of Eq. (A.8) satisfy $a/x^* = 2b(x^*)^2$, so the total is three times the bias term alone,

$$E_L^* = 3 \left\langle \frac{(h_L^*)^2}{12} (I')^2 \right\rangle = \frac{(h_L^*)^2}{4} \bar{\beta}, \quad (\text{B.8})$$

and dividing $G = \langle B(I')^2 \rangle$ by it gives

$$g = \frac{4 \langle B(I')^2 \rangle}{(h_L^*)^2 \bar{\beta}}. \quad (\text{B.9})$$

No input bin width appears in Eq. (B.9). Both G and E_L^* are properties of the profile and of the measurement, so g is independent of the grid the profile was delivered on, as it must be. What it does require is h_L^* , and therefore the noise density of the measurement. Reported uncertainties do not supply that on their own, since they are the variances of the intensity in the bins of the delivered grid rather than a grid-independent noise level, and for a merged or otherwise post-processed reduction the corresponding widths are not always recoverable.

The quantity ρ of Eq. (6) is measurable precisely because it is defined at the delivered grid. For approximately uniform spacing, $\rho = 12 \langle B(I')^2 \rangle / (h_{\text{in}}^2 \bar{\beta})$, and combining this with Eq. (B.9) gives

$$g = \frac{\rho}{3} \left(\frac{h_{\text{in}}}{h_L^*} \right)^2, \quad (\text{B.10})$$

in which the two factors of h_{in} cancel identically, returning Eq. (B.9). Equation (B.10) is therefore a consistency relation and not a route from a measured ρ to a numerical g : refining the grid at fixed measurement increases ρ in proportion to h_{in}^{-2} while leaving g unchanged. Accordingly we report ρ as a diagnostic of the delivered profile, stating whether the instrument or the binning dominates its definition in Q , and we rely on Eq. (B.7) rather than on any numerical value of g for the statement about the verdict, since that statement holds for every $g \geq 0$.

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